

# A dual-aggregation approach to fortify federated learning against poisoning attacks in IoTs

Muawya Al Dalaïen<sup>a</sup>, Ruzat Ullah<sup>b</sup>, Qasem Abu Al-Haija<sup>c,\*</sup>

<sup>a</sup> Cybersecurity Department, Princess Sumaya University for Technology, Amman, Jordan

<sup>b</sup> Department of Computer Systems Engineering, University of Engineering & Technology, Peshawar, Pakistan

<sup>c</sup> Cybersecurity Department, Jordan University of Science and Technology, Irbid, Jordan

## ARTICLE INFO

### Keywords:

Federated learning  
Privacy-preserving  
Poisoning attacks  
Data aggregation  
Internet of Things

## ABSTRACT

Federated learning is gaining much popularity for edge devices. It offers a decentralized approach with strong privacy-preserving capabilities. It has been widely used to secure many edge devices. IoTs also utilize federated learning for an extensive range of security applications. Nevertheless, federated learning itself is also vulnerable to security threats. One such threat is poisoning attacks. Researchers have proposed many models for addressing the issue of poisoning attacks. Most of these approaches come with models based on some external technique (cryptographic or authentication technique), which adds overhead. This paper proposes a dual aggregation approach for securing federated learning. The proposed technique leverages existing machine learning techniques without introducing additional computational overhead. The approach utilizes ensemble learning, where individual client models first aggregate predictions from random forest and gradient boosting, and then the results of all the clients are further aggregated into a global model. Experimental results demonstrate that the proposed method achieves an accuracy of 91 %, highlighting its resilience against model poisoning attacks. The proposed solution provides a lightweight and efficient framework for securing IoT systems, enhancing their resilience against adversarial threats.

## 1. Introduction

Federated Learning (FL) has emerged as a decentralized approach for training across multiple devices or nodes. There has been so much business since the Internet of Things (IoT) surge. Its main advantage is the preservation of data privacy, in addition to decentralization [1]. In FL, Machine Learning (ML) models are trained locally on clients, and after that, these models are aggregated on a central device in a global model, as depicted in Fig. 1. This ensures that the sensitive data remains on local devices [2,3].

FL has gained popularity in domains such as healthcare, finance, edge computing, smart cities, etc., where data security and privacy come at the top [4–6]. Since it is transformative, its data privacy protection comes into play. Its humongous demand can also be attributed to its potential to handle large-scale data while adhering to regulations like GDPR and HIPAA that ensure strict data protection [7]. IoTs are benefiting a great deal from FL. A decentralized FL is viable for applying ML on these nodes since IoTs are spread through nodes and sensors [8]. In recent years, FL has emerged as the most popular solution for IoTs in

cases where ML is applied [9]. This confluence of FL and IoTs provides lightweight solutions, the most pressing demand for power-scarce edge devices.

Like any other viable solution with advantages, it also faces challenges that can hinder its efficacy and threaten its adoption. Some of the recent challenges that FL has been facing include (but are not limited to) system heterogeneity, overhead in communication, and adversarial attacks [10–13]. Among these, poisoning attacks have caught much attention in the research arena due to their mounting threat [14,15]. It can compromise the integrity of FL models. Poisoning attack is rendered in two ways: manipulating the training data by introducing malicious patterns (so that the model trains on wrong data and the time of the actual attack, it is unable to classify it as an ‘attack’), which is also called data poisoning; the other way is model poisoning in which malicious clients send tampered model updates. Both of these attacks can degrade the performance of the global model [16].

Researchers have been attempting to fortify FL and avert any harm to the security of the devices. A. Yazdinejad et al. have proposed a robust model against model poisoning attacks. They have implemented an

\* Corresponding author.

E-mail address: [qsabuhaija@just.edu.jo](mailto:qsabuhaija@just.edu.jo) (Q. Abu Al-Haija).

<https://doi.org/10.1016/j.array.2025.100520>

Received 5 March 2025; Received in revised form 13 August 2025; Accepted 16 September 2025

Available online 19 September 2025

2590-0056/© 2025 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

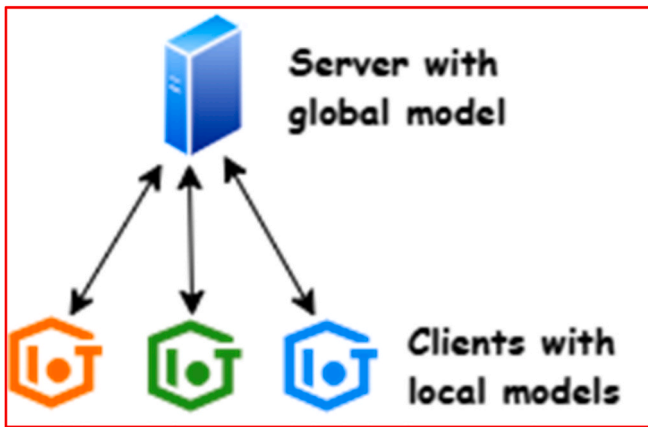


Fig. 1. Working of FL

internal auditor to differentiate between malicious and benign gradients [17]. In another study, S. Nabavirazavi et al. combined aggregation functions. The proposed method switches between three aggregation functions: Switch, Layered Switch, and Weighted FedAvg [18]. In a promising study, A. R Omer et al. have harnessed Trimmed Mean Aggregation and Cosine Similarity Filtering techniques to counteract model poisoning [19]. Researchers have proposed techniques to counter poisoning attacks, including Krum and Trimmed Mean. These proposed solutions still have limitations like increased overhead, communications delay, low accuracy, etc. [20–22]. On the other hand, researchers have widely used EL to secure IoTs. Since EL combines more than one ML model, it is inherently more secure [23]. It offers many advantages over other ML-based models that have been proposed for improving security [24]. EL has been used for industrial IoTs, medical IoTs, anomaly detection, and shielding against attacks [25].

Since EL has proven to be a very pertinent choice of researchers for IoTs, FL has also emerged as a panacea for edge devices. Synergy of both EL and FL can enhance resilience against adversarial threats. This study has explored a novel model for securing the FL. The proposed model leverages the ingrained strengths of Ensemble Learning (EL) to fortify FL against poisoning. The proposed solution integrates EL at both the client and global levels, strengthening security while maintaining efficiency. Deploying EL at both the client and global levels, dual aggregation can be implemented. Enhancing the robustness and reliability of FL improves the overall security of Intrusion Detection Systems (IDSs) in IoT environments. The experiment achieved an accuracy of about 91 %. This research has the following contributions.

- Fortification of FL against poisoning by utilizing the EL at both the client and global levels.
- Utilizing a double aggregation strategy: one at the client level to aggregate the results of individual classifiers (RF and GBM), and the other at the global model to aggregate the client results.
- This approach has exploited an inbuilt security mechanism within FL. This mechanism ensembles two lightweight ML techniques for implementing FL.
- As this technique does not add any exogenous techniques for security (like authentication, cryptography, etc.), no computational overhead is increased, which makes the solution very suitable for an IoT environment.
- Two different aggregation mechanisms are used to ensure more robustness. If one of the aggregations is compromised, the other could still ensure resilience.
- The proposed model has demonstrated an accuracy of 91 % while maintaining computational efficiency simultaneously. It makes the model a viable solution for real-world IDS.

The rest of the paper is organized in such a way that section 2 discusses background concepts used in this research; section 3 reviews the relevant literature; section 4 is about data acquisition, data pre-processing, and features selection; section 5 discusses the methodology of the research; results have been discussed in section 6; in section 7, the paper has been concluded with future directions.

## 2. Background

Before going into the details of the experiment, it is pertinent to first delve into the prerequisite concepts used in this research. Among these concepts are FL, EL, and Poisoning attacks. This section will shed light on the techniques used in this experiment.

### 2.1. Federated learning

In the classic ML mechanism, we collect data and build a model. However, unlike orthodox ML models, the model is applied to the data in FL. It provides a decentralized way of building ML models. There are several nodes/clients<sup>1</sup> and a server [26]. The server has a global model, while the nodes have local models. Local models send their updates to the global model, which aggregates these updates and averages them. This results in an improved global model [27]. FL is very helpful in preserving the privacy of sensitive data. Furthermore, its collaborative training nature facilitates the building of more accurate models [28].

### 2.2. Ensemble learning

In EL, various ML models are combined to predict the classes more accurately. In this mechanism, the various models train individually on the data and then combine their predictions to improve accuracy and other metrics. The results/predictions are combined using different

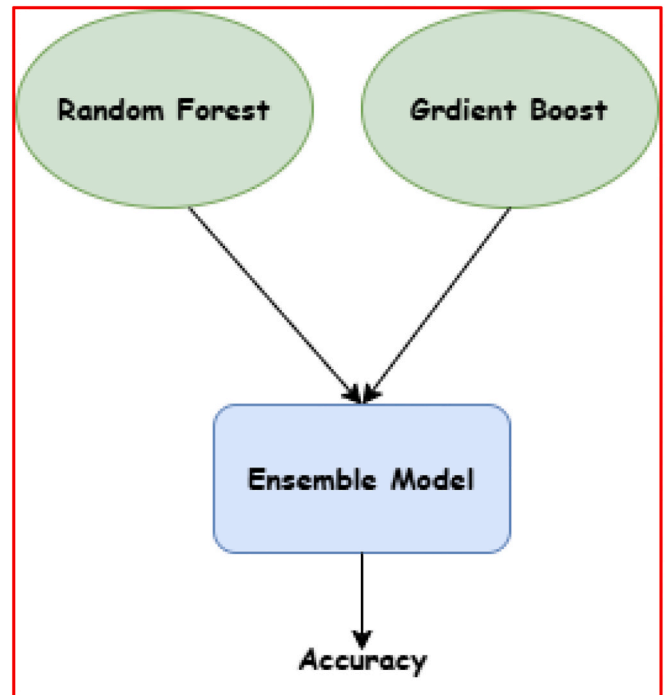


Fig. 2. Working of EL

<sup>1</sup> Node and client are used interchangeably in FL.

techniques like voting, averaging, or weighting [29]. The process of EL is depicted in Fig. 2. EL is very popular for IoT security cases [30–32].

### 2.3. Poisoning attacks

Poisoning attacks are attacks in which the attacker intentionally sends malicious model updates or training data to the global model. This prevents the global model from converging and hence hinders accurate predictions. Although poisoning attacks were first proposed by Battista et al. long before the advent of FL, with the surge of FL, these attacks also became more prevalent [14]. Poisoning attacks can be one of these two types: data poisoning and model poisoning. In the case of the former, the training data is manipulated (like flipping the features), while malicious clients send corrupt updates to the global model in the case of the latter.

## 3. Literature review

Researchers have been working meticulously to make the technology better and secure. The research arena has successfully averted many of the challenges upfront for FL. Similarly, experts have been unearthing new ways of securing IoTs. EL is one such approach that is reputed for securing IoTs. Since this research incorporates FL and EL, this literature review section will have two parts: the first will discuss state-of-the-art literature on securing FL against poisoning attacks, and the second will be about using EL as a successful security model for IoTs.

### 3.1. Securing FL against poisoning attacks

S. B. Saad et al. have proposed a novel model for securing FL against model poisoning attacks in 5G and beyond networks. Their simulations were specifically for mobile devices. They have utilized the concept of Zero Touch Management (ZTM) for security. Specifically, the security mechanism involves two steps: first, a trusted participant is selected dynamically based on its record; second, this selected trusted participant then detects malicious model updates. Deep reinforcement was used to select trusted participants, while unsupervised learning was used to detect poisoning updates [33]. Meanwhile, the DefendFL approach has been put forward by J. Liu et al., which is a privacy-preserving scheme for FL against poisoning attacks. A collinearity mask was used to protect users' gradients. Cosine similarity was used to detect poisoned gradients. As another level of security, a verification mechanism has been integrated to detect the mask. Finally, the malicious gradients are detected and removed, or their weight is decreased in the model [34].

In another paper by J. Zhou et al., FL has been preserved and secured against privacy leakages and poisoning attacks. A differential privacy technique has been used to preserve the privacy of the user data and the model. For securing FL against poisoning attacks, a weight-based anomaly detection model has been proposed to detect any malicious features if uploaded by nodes. This approach combines privacy preservation and fortification against model poisoning simultaneously. Results show that this approach has outperformed other approaches in the literature [35].

ApFed has been introduced by X. Chen et al. in their article. The problem of securing FL against poisoning in a heterogeneous data scenario has been addressed in this research. The proposed solution uses homomorphic encryption to enhance privacy preservation, while a cosine similarity scheme has been used to detect poisonous gradients. A clustering mechanism has also been incorporated for hierarchical aggregation. Overall, it has been shown experimentally that the proposed solution performs quite better than the earlier solutions as it mitigates the attacks and reduces communication overhead [36].

Z. Zhang et al. have explored yet another such model called SecFedNIDS. This is a robust and secure FL-based network intrusion detection system. The proposed model is twofold: firstly, model-level defense has been performed using gradient-based important feature selection, and then an online unsupervised detection model eliminates

the need for a pre-trained offline model; secondly, for data-level defense, layer-wise relevance propagation has been used to extract class paths from cleaned traffic. It also filters out poisoned data and retrains the local model with clean data. It has shown a maximum of 36 % enhancement in accuracy on the UNSW-NB15 dataset [37].

The research findings on securing Federated Learning (FL) against poisoning attacks are summarized in Table 1. This table presents a comparative analysis of various approaches, outlining their methodologies, defense mechanisms, and effectiveness in mitigating adversarial threats.

### 3.2. Using EL and FL as viable security models for IoTs

S. Zawad et al. have come up with FedCust. This is an FL-based approach to overcome the power-consuming hyperparameter tuning. In this approach, the customization of hyperparameters has been off-loaded to servers instead of clients. Further, the research has explored that not all clients need customization of the hyperparameters. Clients with the same heterogeneous data require the same hyperparameters. On all the popular FL benchmarks, FedCust has achieved much better accuracy [38].

In a promising research, D. Jayalatchumy et al. have proposed a three-step solution for an IoT Intrusion Detection System (IDS). In the first step, data imbalance has been removed using denoising methodologies. In the next step, the Crow search algorithm has been used to select optimal features for better intrusion classifications. Lastly, four distinct classifiers have been ensembled for classification. For evaluation, NSL-KDD and UNSW-NB15 were used in their denoised versions. Results demonstrated that 99.4 % and 99.2 % accuracies were achieved, respectively [39].

N. Jeffery et al. have unearthed a hybrid anomaly detection approach. In this approach, the authors combined classic signature-based and ML-based anomaly detection. The proposed model assigns detection and classification of some threats to a signature-based and threshold-based mechanism, which uses domain knowledge and is computationally very inexpensive. Besides this, some of the threats have been assigned to ML-based classification. For ML-based classification, the EL approach has been used. This proposed mechanism increases accuracy while decreasing computational cost. Results have shown that it has achieved a 4–7 % accuracy improvement over the traditional ML classifiers when evaluated on the same publicly available datasets [40].

A novel aspect of intrusion detection has been discussed by M. Osman et al. in their research. The authors have proposed an EL-based approach for intrusion detection against a particular routing protocol called RPL. In this approach, feature extraction and selection have been performed using a genetic algorithm. Three RPL attack classes have been

**Table 1**  
Summary of research on securing FL against poisoning attacks.

| Reference         | Issue Addressed                                                                                      | Limitation                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| S. B. Saad et al. | Model poisoning using unsupervised learning and deep reinforcement learning                          | Selecting trusted participants is still a vulnerable solution, as it can also be compromised.     |
| J. Liu et al.     | Detecting malicious gradients against benign gradients using the masked mechanism                    | This can result in increased overhead on the edge devices, which is not wanted.                   |
| J. Zhou et al.    | Privacy preservation of user data and model. Besides that, the poisoning attack on FL also addressed | This is a viable approach, but model poisoning is not addressed, which can still be problematic.  |
| X. Chen et al.    | Securing FL in the scenario of heterogeneous data                                                    | It is very generic in securing FL, whereas the poisoning has become very sophisticated.           |
| Z. Zhang et al.   | It has incorporated both models as well as data-level fortification against poisoning.               | The solution will be resource-costly because new techniques have been added, increasing overhead. |

used: version number, decreased rank, and DIS flooding. EL was used for classification. An accuracy of 97.90 % has been recorded for multiclass classification [41].

H. Fares et al. have proposed an interesting model for combating intrusions in IoTs. In the proposed solution, three different ML models have been used, including Deep Neural Network (DNN), Decision Tree (DT), and Random Forest (RF). Later, they have also introduced EL. The main classes of attacks discussed are DoS (Denial of Service) and DDoS (Distributed Denial of Service). Experiments were carried out on CICIOT-2023 and WSN-DS datasets. It was evident from the evaluation that EL was much more effective than the singular ML models [42].

The research findings on employing Edge Learning (EL) and Federated Learning (FL) as viable security models for IoT environments are summarized in Table 2. This table provides a comparative analysis of various studies, highlighting their methodologies, key contributions, and security implications.

#### 4. Dataset acquisition

As a prerequisite for running, the dataset must be obtained first. Many publicly available datasets are widely used for FL in IoTs. These datasets include, but are not limited to, MNIST (and FEMNIST), UNSW-NB15, KDD Cup 1999, CIFAR-10/100, NSL-KDD, UCI HAR, CICIOT-2023, and IoT23 dataset, etc. [43–46]. Among these datasets, we selected CICIOT-2023 as public, latest, large, popular, and incorporating 33 attacks. Click or tap here to enter text. These attacks are classified into seven distinct categories. This dataset includes almost all the important features for training ML models for IoT security. The dataset is available in PCAP and CSV (Comma Separated Values). Since CSV suited our experiment more, we downloaded it in CSV format.

##### 4.1. Preprocessing

CICIOT-2023 is a very large dataset stored in over 60 separate files. We gradually increased the amount of data to observe its impact on results. The best results we achieved were when selecting one million instances. Besides that, the preprocessing also included handling missing values and encoding target classes. One important aspect of the preprocessing included balancing the dataset. The selected dataset has an imbalanced distribution of the classes, which hinders the performance. Five important classes were filtered out to address this using ADASYN (Adaptive Synthetic Sampling). ADASYN generates synthetic samples for the minority class using Eq. (1) formula.

$$\mathbf{x}_{new} = \mathbf{x}_i + \lambda \cdot (\mathbf{x}_{nn} - \mathbf{x}_i) \quad \text{Equation 1}$$

Here,  $\mathbf{x}_i$  is an instance of the minority class,  $\mathbf{x}_{nn}$  is one of its nearest

neighbours, while  $\lambda$  is a random value between 0 and 1. Furthermore, the dataset was divided into training and testing sets with an 80 %–20 % ratio. The `train_test_split` function was used to do this.

##### 4.2. Feature selection

We used Random Forest (RF) in this research to select optimal features. The algorithm first evaluates the performance of different combinations of features and then selects the most influential features. Based on these algorithms, a total of 15 features were selected. These features are selected based on their importance score, while feature importance is determined using the decrease in impurity. Mathematically, it can be shown in Eqs. (2) and (3).

$$\Delta I(\mathbf{f}, \mathbf{t}) = I(\mathbf{t}) - \left( \frac{N_{left}}{N} I(\mathbf{t}_{left}) + \frac{N_{right}}{N} I(\mathbf{t}_{right}) \right) \quad \text{Equation 2}$$

$$\text{Importance}(\mathbf{f}) = \frac{1}{T} \sum_{j=1}^T \sum_{t \in T_i} \Delta I(\mathbf{f}, \mathbf{t}) \quad \text{Equation 3}$$

Here,  $T$  is the total number of trees,  $\mathbf{t}$  is used for nodes of feature  $\mathbf{f}$ ,  $I$  is an impurity.

#### 5. Methodology

In this research, fortification of FL against model poisoning attacks has been proposed. This has been performed using EL. This is a robust framework where FL integrates EL in both the global and client models. Here is how the model's working has been proposed in Fig. 3.

##### 5.1. Data poisoning

To add poisoned data, 10 % of the training dataset was randomly selected. This selected data was perturbed with noise. We selected the Gaussian noise approach for adding noise, as shown in Eq. (4). Here,  $\mathbf{X}$  represents the feature matrix.

$$\mathbf{X}_{poisoned}[\mathbf{i}, \mathbf{j}] = \mathbf{X}[\mathbf{i}, \mathbf{j}] + \epsilon, \epsilon \sim \mathbf{N}(0, 1) \quad \text{Equation 4}$$

These features' class values (represented by  $\mathbf{y}$ ) are also replaced by random class values, as shown in Eq. (5).

$$\mathbf{y}_{poisoned}[\mathbf{i}] = \text{random}(\mathbf{y}) \quad \text{Equation 5}$$

The data poisoning step reflects the adversarial challenge as it should arise in the real-world FL. This simulation imitates the behaviour of malicious clients in FL systems. It is also worth-mentioning that the poisoning was added to the datasets before the training. This happened to deteriorate performance of the clients which is an actual manifestation of poisoning attack.

##### 5.2. Data distribution

As it has already been discussed that the dataset is divided in the proportion of 80 % and 20 % for training and testing respectively. The training part of the dataset is equally and equitably distributed among five clients. Each of the clients gets a 20 % share of the clients' dataset. For attack simulation, 10 % of the training data is randomly selected to be poisoned. The data distribution is Independent and Identically Distributed (IID). The data distributed under IID is in such a way that all the classes of the original dataset are distributed among all the clients in almost the same proportion as in the original dataset.

##### 5.3. Integration of EL with FL

The FL setup was simulated with five attached clients. The training data was distributed equally among these clients. Each of the clients was trained on its own dataset. The poisoning was realized by adding

**Table 2**  
Summary of research on using EL and FL as viable security models for IoTs.

| Reference              | Issue Addressed                                                                                               | Limitation                                                                                     |
|------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| F. Zawad et al.        | Power-hungry fine-tuning of hyperparameters offloaded to a server                                             | Security issues can still arise that have not been addressed                                   |
| D. Jayalatchumy et al. | Data imbalance was removed, optimal features were selected, and EL was applied for classification.            | Feature selection can still be affected by poisoning.                                          |
| N. Jeffery et al.      | Anomaly detection in a hybrid manner by combining ML and signature-based methods reduces computational costs. | Privacy preservation can be a big issue in this case.                                          |
| M. Osman et al.        | RPL attacks have been averted using EL                                                                        | There are a lot more attack classes than RPL                                                   |
| H. Fares               | Securing IDS against DoS, DDoS, and their offspring using EL                                                  | It can be computationally costly, as well as many other types of attacks, that are unaddressed |



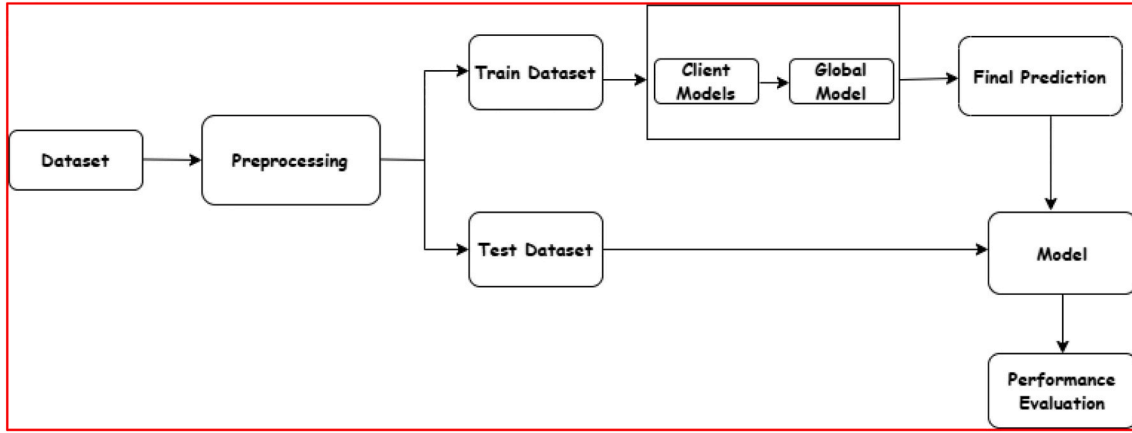


Fig. 3. Methodology of the research.

Gaussian noise to the nodes. 10 % of each of the client's dataset was poisoned. That is, all of the clients consisted poisoned data.

The clients were connected to a central device called server. The server has the global model, while clients have node models. Each of these six entities utilized EL for the implementation of FL. EL includes two very lightweight ML models (fit for power-scarce IoTs): RF and Gradient Boosting Model (GBM). RF and GBM have been chosen because.

- RF can learn high variance effectively.
- RF can learn general patterns well, too.
- GBM is good at minimizing the error iteratively
- The problem of overfitting doesn't normally occur in GBM

RF has already been explained mathematically in the Background section. At the same time, GBM is mathematically shown in Eq. (6). This equation shows the iterative updating process of GBM, where it keeps adding a new weak learner.  $h_m(x)$ . The learning rate then scales.  $\nu$ .

$$F_m(x) = F_{m-1}(x) + \nu \cdot h_m(x) \quad \text{Equation 6}$$

These two classifiers are trained separately, and their result is aggregated using a soft voting mechanism, as shown in Fig. 4. Soft voting combines the results obtained from the classifiers individually and selects the target class with the highest probability. The predicted class is the one with a higher average.

Mathematically, the soft voting mechanism of the ensemble model at nodes first combines the probabilistic predictions from the RF and GBM, as in Eq. (7):

$$P_{ensemble}(y = c|x) = \frac{1}{2} (P_{RF}(y = c|x) + P_{GBM}(y = c|x)) \quad \text{Equation 7}$$

After combining these individual probabilities, the class with the

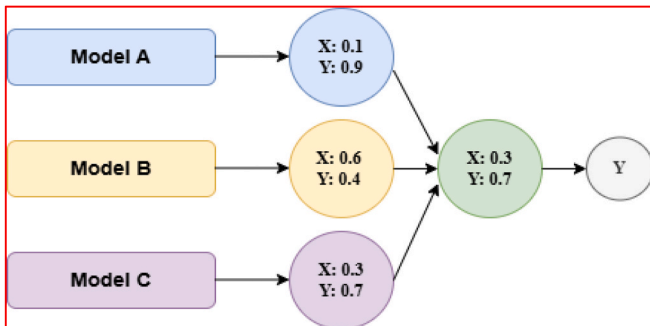


Fig. 4. Soft voting mechanism.

highest probability is selected, as in Eq. (8):

$$\hat{y} = \underset{y}{\operatorname{argmax}} P_{ensemble}(y = c|x) \quad \text{Equation 8}$$

When all these node models get trained, they send their predictions to the global model. The global model averages the outputs from all the clients ( $N$ ), as shown in Eq. (9).

$$P_{global}(y = c|x) = \frac{1}{N} \sum_{i=1}^N P_{client_i}(y = c|x) \quad \text{Equation 9}$$

The averaging of the predictions from all the models minimizes the adverse effect of poisoned predictions received from a model. Since the averaging method of aggregation holds the majority of the weight.

#### 5.4. Dual aggregation scheme

The aggregation proposed in this model is two-folded: one at the client-level and other at global server level. At the client side, two separate models, RF and GBM, have been employed for predictions. The results of these individual classifiers are aggregated at each client using soft voting. The aggregated results from all the clients are then sent to the global model. At the global model, the results received from the individual clients are aggregated using averaging. The global model after aggregating the clients' results, is subjected to the test dataset. The test dataset is used to evaluate the global model.

#### 5.5. Security model

The security model deployed in this research works with the global model and the clients. Global model, which is at the server, is an honest central device. Its only aim is to aggregate the updates received from the clients. The data contained in it is clean and there is no malicious perturbation in it. On the other hand, all of the clients are partially malicious. Clients have malicious data instance at features as well as at labels. Each of the clients has 10 % of its data as malicious.

#### 5.6. Experimental setup

Python was considered a programming language for its versatility when running this experiment. All the experiments were run on the Google Colab engine [47]. A lot of Python libraries have been utilized to carry out this experiment. Some of these libraries, with their brief description, are as follows.

- **Numpy:** It is used for handling multi-dimensional arrays of matrices. It makes the dataset ready for applying ML models [48].

- **Pandas:** This is a very powerful Python tool. It is mainly used to import and clean the dataset and handle missing values [49].
- **Scikit-learn:** It is the main library used. The ML side of the experiment has been implemented using Scikit-learn. From EL to FL, all the steps have been performed using this tool. Many scikit-learn modules, like base, clone, ensemble, and model\_selection, have been imported and used in this experiment [50].
- **Matplotlib:** This library is an extension of Numpy. It is used for drawing plots and graphs, etc. [51].
- **Seaborn:** This Python library has been built on top of Matplotlib. It is used for more meticulous and aesthetically sophisticated graphs [52].

### 5.7. Attack model

A poisoning attack was simulated before the training phase to evaluate the resilience of our proposed model. Adversarial noise was introduced into the data to carry out this simulation. Training on noisy data is aimed at degrading the performance of client models, which in turn degrades the performance of the global model. For poisoning, a two-step approach has been adopted.

#### 5.7.1. Poisoning strategy

The poisoning strategy mainly involves modifying two parts of the data.

- **Manipulating features:** Gaussian noise (with  $\mu = 0$ ,  $\sigma = 0.5$ ) has been added to the data feature values, simulating adversarial corruption.
- **Flipping labels:** The class labels of poisoned data have also been randomly replaced with some incorrect class labels. This mimics a targeted poisoning attack.

#### 5.7.2. Injection of poisoned data

10 % of the data is randomly selected when poisoned data is injected into the original dataset. The poisoned data replaces the selected data. Thus, injection of the poisoned data simulates a real-world scenario.

### 5.8. Algorithm

The proposed solution works this way.

Algorithm: Dual Aggregation Federated Learning (DAFL)

1. Split the dataset  $D$  in two sets: training ( $D_{\text{train}} = 80\%$ ) and testing ( $D_{\text{test}} = 20\%$ )
2. Introduce poisoning in  $D_{\text{train}}$ 
  - a. Select a 10 % of  $D_{\text{train}}$
  - b. Add Gaussian noise and flip the labels
3. Divide  $D_{\text{train}}$  into five clients  $D_1, D_2, D_3, D_4, D_5$
4. For each client in clients:
  - a. Train RF and GBM at each client
  - b. Aggregate results of RF and GBM using soft voting for ensemble learning ( $EL_i$ )
5. Send the ensemble models ( $EL_i$ ) to global model  $M_G$
6. Aggregate results from all the clients at global server
7. Evaluate performance using:
  - a. Accuracy, precision, recall and f1-score
  - b. ASR (for poisoned data) and benign accuracy (for non-poisoned data)

## 6. Results and discussion

Results come as a demonstration of the validation of the proposed model. Since the proposed approach uses EL to secure the FL, the proposed model has no measures taken out of the domain of ML. This is why, for this experiment, the primary results we have considered are accuracy, Precision, Recall, and f1\_score. These metrics are the popularly accepted benchmark metrics for evaluating an ML-based solution. ASR and benign accuracy have been considered as additional metrics which are FL security specific.

- **Accuracy:** It measures how well a classifier has performed the classification.
- **Precision:** This metric measures the percentage of correctly made predictions.
- **Recall:** It shows the correct identification of relevant data points by the model.
- **F1 Score:** It combines the values of Precision and Recall by taking their harmonic mean.
- **Attack Success Score (ASR):** One of the popular metrics used specifically for poisoning attacks on FL is ASL. It is actually the percentage of the misclassified poisoning attacks. If the global server classifies a poisoning instance as non-poisoned, this is called attack success rate. The lower value of ASR is better.
- **Benign Accuracy:** The ratio of non-poisoned (clean) data that is rightly classified as clean by the global model is called benign accuracy.

The experiment was performed in two tiers: the first tier was at the node level, and the second one was at server level. At the node level, the individual models (RF and GBM) were trained (using training data) and their results were aggregated using soft voting. At the server level, the global model aggregated the results of all the clients. The global model was then tested with the 20 % of the test dataset.

It is a matter of fact that the performance metrics of a model degrade after being afflicted by poisoning. Nevertheless, the proposed technique has yielded an accuracy above 91 % for its global model, as seen in Fig. 5. It is of commendable accuracy.

Similarly, the Precision of this model is also above 90 % for its global model. This shows that very correct predictions have been made. In addition, the Recall score is 91 %, while the F1 score is 90 %. All of these measures have values that are very close to each other because they were obtained using 'micro' averaging.

Besides accuracy, precision, recall, and F1-score, one important measure of FL models is the time cost. The model that consumes more time is generally considered more complex, while the model that takes less time is considered less complex. Increasing complexity often results in more power consumption. Conversely, IoTs can hardly accept any power-consuming solution. The proposed model took 1430.001195 s, which equals 23.833 min. This is a commendable degree of trade-off between accuracy and time cost.

As far as the client models are concerned, they have also performed very well. Despite being trained on 10 % poisoned data, they still have accuracies of around 80 %, and similar are their other metrics. Their results can be seen in Fig. 6.

It can be inferred from comparing the global model with client models that the client models have deteriorated results because of the introduction of poisoned data. FL has successfully filtered out this degradation of the result. In client models, the highest accuracy has been achieved by client 3, which is 79.95 % (almost 80 %), whereas the global model has an accuracy of 91 %. Regarding Precision, the same client 3 has the highest performance, 79.93 %, whereas the global model has a precision of 90 %. Regarding Recall, 79.95 % (almost 80 %) compared to the global model, which is 91 %. Similarly, the same client 3 has the highest F1-score, 79.92 %, whereas the global model has an F1-score of 90 %, as seen in Fig. 7. Here, the best-performing client is client 3.

The results are comprehensively summarized in Table 3 below, providing a clear and structured overview of the key findings. This table highlights the most relevant performance metrics, enabling a comparative analysis of the results.

For further strengthening the claims of this research, ASR and benign accuracy have also been measured. The ASR stood at 0.07 which is almost about 7 %. The low value of ASR shows the robustness of the model. Similarly, benign accuracy was 0.91 which is almost 91 %. An overview of global model metrics including ASR and Benign Accuracy Rate (BAR) has been given in Fig. 8.

To draw a comparative analysis, we've run another EL. The model

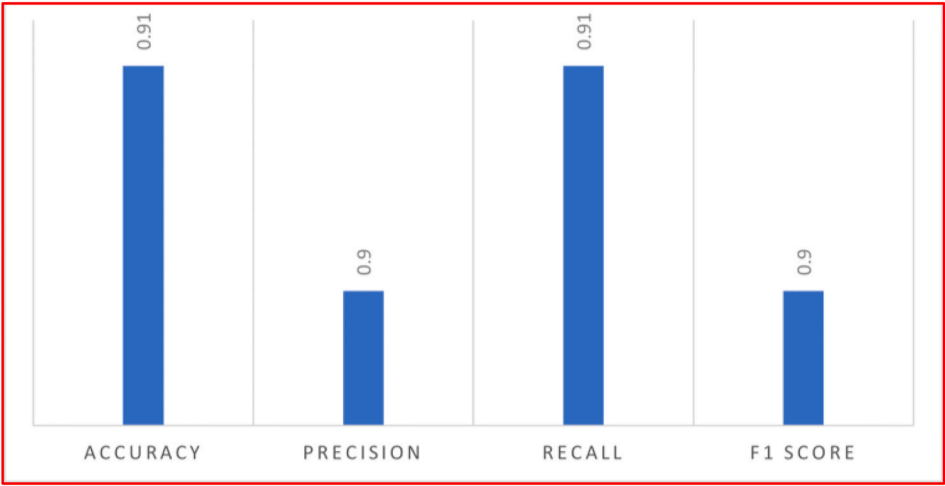


Fig. 5. Performance of global model.

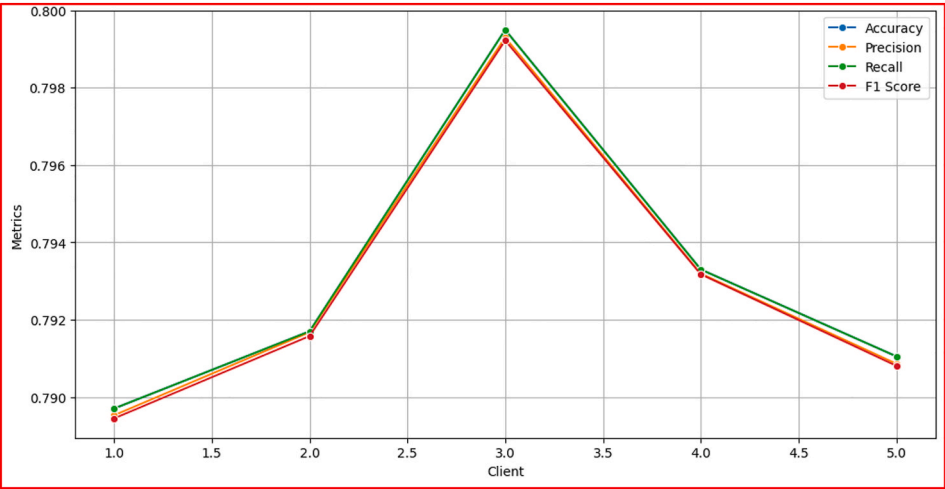


Fig. 6. Performance of client models.

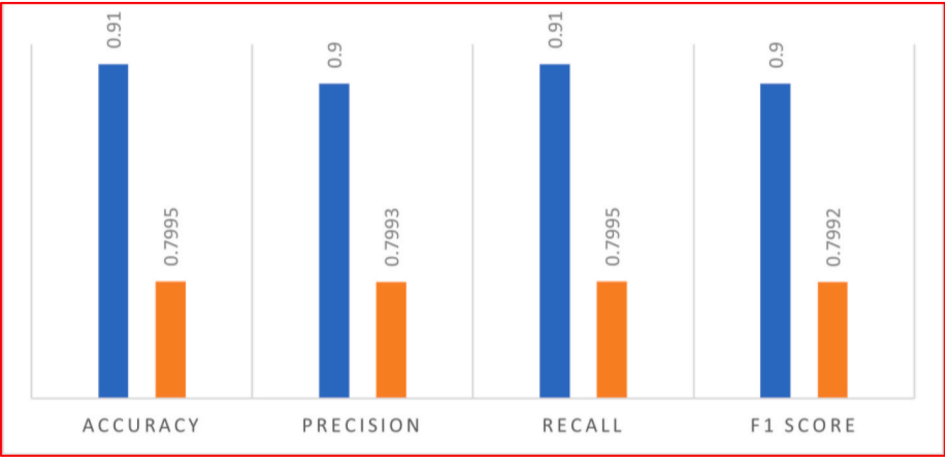


Fig. 7. Global Model vs Best Performing Client (client 3).

for comparison has an ensemble Logistic Regression (LR) model and a Naïve Bayes (NB) model. The comparison is aimed at demonstrating the effectiveness of the proposed solution. It is evident from the results that our proposed model has achieved much better performance in terms of

all the metrics. The metrics show that EL with LR and NB achieved much lower accuracy than EL with RF and GBM. The former has achieved an accuracy of 77 % whereas the latter 91 %. Similarly, regarding precision, the

**Table 3**  
Performance of client models.

| Client | Accuracy | Precision | Recall  | F1-score |
|--------|----------|-----------|---------|----------|
| 1      | 0.78970  | 0.789526  | 0.78970 | 0.789445 |
| 2      | 0.79170  | 0.791677  | 0.79170 | 0.791577 |
| 3      | 0.79950  | 0.799301  | 0.79950 | 0.799225 |
| 4      | 0.79330  | 0.793193  | 0.79330 | 0.793174 |
| 5      | 0.79105  | 0.790860  | 0.79105 | 0.790806 |

former gave a value of 77 % while the latter (as already discussed) gave 90 %. Regarding recall, our proposed approach achieved a value of 91 % while EL with LR and NB achieved 77 %. In prediction performance, the F1-score of the proposed approach to EL with LR and NB is 90 %–77 % respectively. Moreover, one of the important aspects in every model is its time cost. It is especially important in IoTs. A model with time cost and accuracy trade-off is often recommended. The approach has a time cost of 1430.001195 s while the comparative model has a time cost of

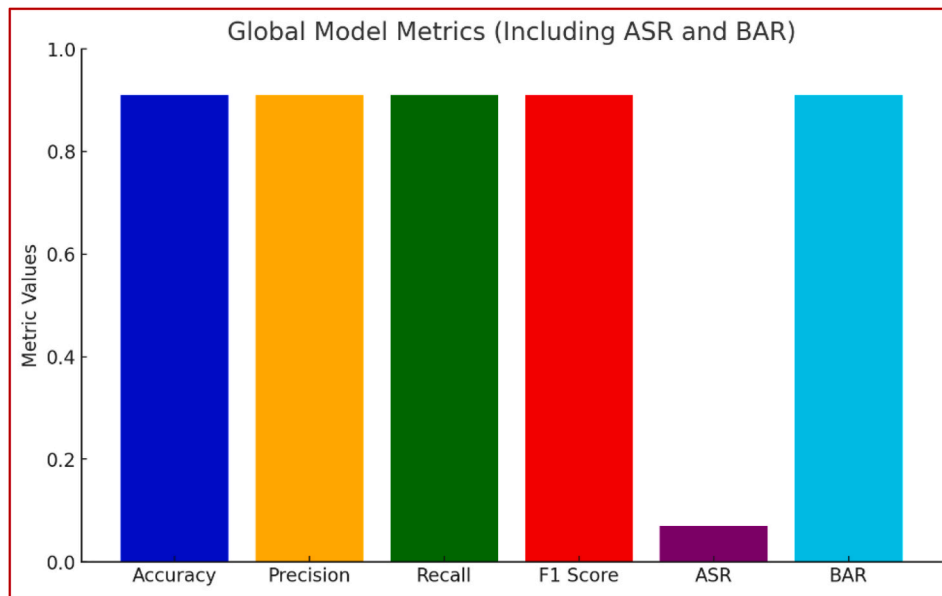
544.48 s. A comparative analysis of both models is given below in Fig. 9 and Table 4.

The results show that the global model of this EL-based technique has proven to be very robust and resilient against poisoning attacks. Even in the face of enough poisoning, its performance could not deteriorate. It has shown high performance against all the metrics, which is a testimony to its robustness. The results also demonstrated that the proposed model is accurate and efficient regarding time consumption. This is proof of the viability of the proposed model when adopted.

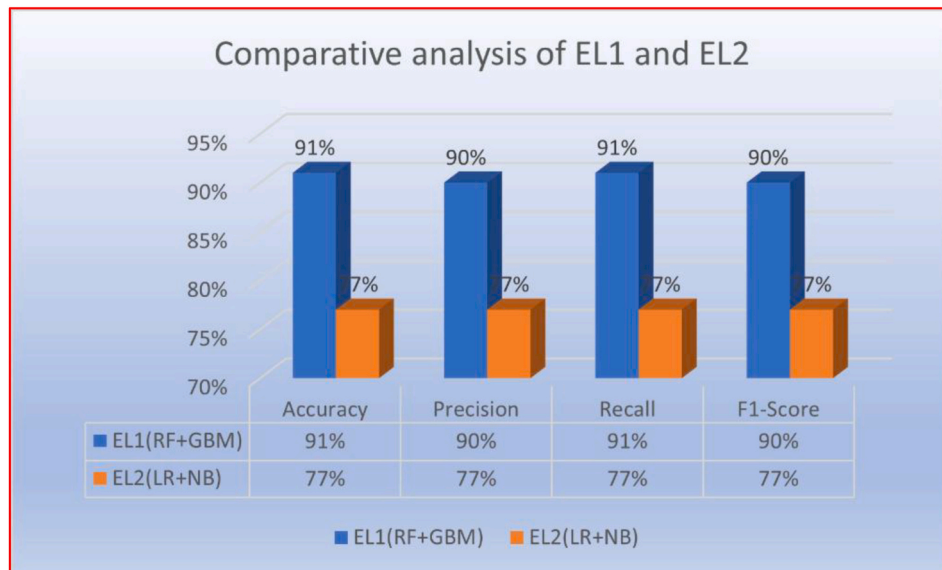
Moreover, we have also run the same experiment with non-IID data

**Table 4**  
Comparison of time cost.

| Model          | Time cost (in seconds) |
|----------------|------------------------|
| EL1 (RF + GBM) | 1430.001195            |
| EL2 (LR + NB)  | 544.48                 |



**Fig. 8.** Global model metric including ASR and benign accuracy.



**Fig. 9.** Comparison of EL1 and EL2.



(as in many cases of FL). The results show that the global model resulted in almost the similar performance as that of IID data. The global model with non-IID data gave an accuracy and other metrics of 90 % which are almost the same as IID. The global model results have been shown in Fig. 10. The performance of the client models derailed a lot in non-IID data. Their accuracy and other metrics stood around 35 % as shown in Fig. 11. It is also worth-mentioning here that many IoT devices generate IID data like in case of a group of sensors, all recording temperature or tracking activities. In such cases, non-IID data may not even be needed. Besides that, in our current setting, the global model of non-IID has almost the similar performance as that of IID, so both the settings can be equally applied (as global model is the most important aspect in FL and has the highest influence).

There relevant works in the literature that closely resemble this research. Comparison of our results with those works can better highlight the strengths and weakness of this research. The most relevant research in this direction has been carried by S. B. Prathiba et al. The authors have introduced a blockchain and digital twin-based solution for securing against poisoning attacks. Their proposed system has achieved an accuracy of 97 % but it results in additional overhead through digital twins and block-chain based verification. Additionally, they also haven't taken into account the aspects like ASR [53]. Moreover, M. Wasilewska et al. have investigated the detection of poisoning attack on FL in spectrum sensing in case of cognitive radio communication. The attacker models have been detected with probabilities ranging from 90 % to 100 %. However, the detection model used is not as lightweight as that of used here. Along with that, the global model employs continuous monitoring of the clients which can result in very frequent communication and hence causing many issues [54]. In another very relevant work, Shashidhar R et al. have proposed a defense mechanism against poisoning attacks in 5G and beyond networks. The proposed approach, DBFFL, has shown resilience against poisoning attacks with an accuracy of 91.6 % when tested on MNIST dataset. One main shortcoming in the research is that it only addresses poisoning attacks for mobile networks [55]. A solution named DDPG has been put forward by T. Wan et al. to select fair clients in FL. This fair selection of clients ensures secure FL setting. The accuracy value of DDPG on the MNIST is 0.941 and 0.923. Since the model selects clients based on security-fairness values which make it prone to errors [56].

## 7. Conclusion and future work

FL has become very popular in many IoT applications because of its decentralization. Despite the numerous advantages of IoT, it still has to face many challenges. One such challenge that FL faces is Poisoning attacks. Many attempts have been made to combat these challenges. Although these proposed approaches have some benefits, they mostly lack versatility. In this research, we've proposed an EL-based model for overcoming poisoning attacks in IoTs. The proposed solution is a lightweight, robust solution. Instead of adding any exogenous technique, ML was used to fortify FL. The proposed solution uses dual aggregation: one in the case of EL at the client model and the other in the case of global model aggregation using an averaging technique. This makes the proposed technique very secure against poisoning attacks. The results have been well-validated by the given results. In the future, we aim to extend the scope of this proposal further to test an ensemble of deep learning models. This can lead us to more accurate results.

## CRediT authorship contribution statement

**Muawya Al Dalaïen:** Writing – review & editing, Writing – original draft, Resources, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ruzat Ullah:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Qasem Abu Al-Haija:** Writing – review & editing, Writing – original draft, Visualization, Validation,

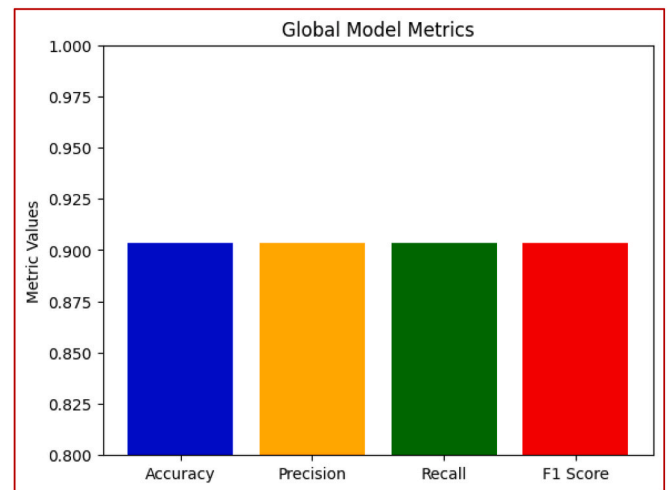


Fig. 10. Global Model Metrics in non-IID data.

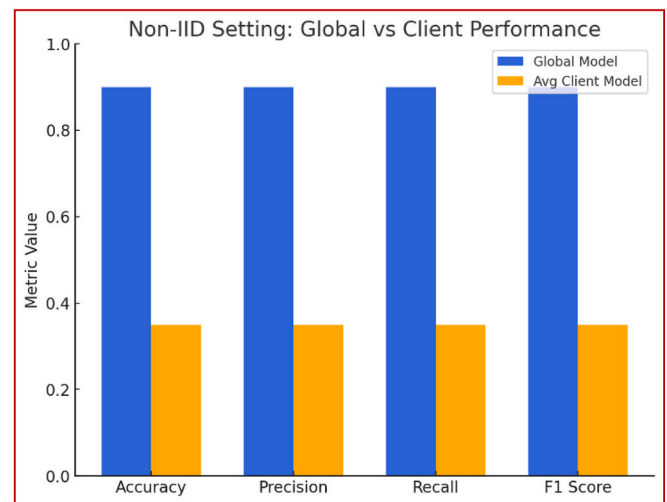


Fig. 11. Global Model vs Client Model.

Investigation, Formal analysis.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## References

- [1] Jiang JC, Kantarci B, Oktug S, Soyata T. Federated learning in smart city sensing: challenges and opportunities. *Sensors* Oct. 2020;20(21):6230. <https://doi.org/10.3390/s20216230>.
- [2] Antunes RS, André da Costa C, Küderle A, Yari IA, Eskofier B. Federated learning for healthcare: systematic review and architecture proposal. *ACM Trans Intell Syst Technol* Aug. 2022;13(4):1–23. <https://doi.org/10.1145/3501813>.
- [3] Nguyen DC, et al. Federated learning for smart healthcare: a survey. *ACM Comput Surv* Mar. 2023;55(3):1–37. <https://doi.org/10.1145/3501296>.
- [4] Brauneck A, et al. Federated machine learning, privacy-enhancing technologies, and data protection laws in medical research: Scoping review. *J Med Internet Res* Mar. 2023;25:e41588. <https://doi.org/10.2196/41588>.

- [5] Ghimire B, Rawat DB. Recent Advances on federated learning for Cybersecurity and Cybersecurity for federated learning for Internet of Things. *IEEE Internet Things J* Jun. 2022;9(11):8229–49. <https://doi.org/10.1109/JIOT.2022.3150363>.
- [6] Zhang T, He C, Ma T, Gao L, Ma M, Avestimehr S. Federated learning for Internet of Things. In: *Proceedings of the 19th ACM Conference on Embedded networked sensor systems*. New York, NY, USA: ACM, Nov.; 2021. p. 413–9. <https://doi.org/10.1145/3485730.3493444>.
- [7] Lin S, Zhai D, Zhang F, Jiang J, Liu X, Ji X. Overhead-free noise-tolerant federated learning: a new Baseline. *Mach Intell Res Jun*. 2024;21(3):526–37. <https://doi.org/10.1007/s11633-023-1449-1>.
- [8] Chai Z, et al. TiFL: a tier-based federated learning system. In: *Proceedings of the 29th International Symposium on high-performance Parallel and distributed computing*. New York, NY, USA: ACM; Jun. 2020. p. 125–36. <https://doi.org/10.1145/3369583.3392686>.
- [9] Wu W, He L, Lin W, Mao R, Maple C, Jarvis S. SAFA: a semi-asynchronous protocol for Fast federated learning with low overhead. *IEEE Trans Comput May* 2021;70(5):655–68. <https://doi.org/10.1109/TC.2020.2994391>.
- [10] Zhang Y, et al. A survey of Trustworthy federated learning: issues, solutions, and challenges. *ACM Trans Intell Syst Technol Dec*. 2024;15(6):1–47. <https://doi.org/10.1145/3678181>.
- [11] Xia G, Chen J, Yu C, Ma J. Poisoning attacks in federated learning: a survey. *IEEE Access* 2023;11:10708–22. <https://doi.org/10.1109/ACCESS.2023.3238823>.
- [12] Nguyen TD, Nguyen T, Le Nguyen P, Pham HH, Doan KD, Wong K-S. Backdoor attacks and defenses in federated learning: survey, challenges and future research directions. *Eng Appl Artif Intell Jan*. 2024;127:107166. <https://doi.org/10.1016/j.engappai.2023.107166>.
- [13] Wan Y, Qu Y, Ni W, Xiang Y, Gao L, Hossain E. Data and model poisoning Backdoor attacks on Wireless federated learning, and the defense mechanisms: a comprehensive survey. *IEEE Commun Surv Tutor* 2024;26(3):1861–97. <https://doi.org/10.1109/COMST.2024.3361451>.
- [14] Yazdinejad A, Dehghantanha A, Karimipour H, Srivastava G, Parizi RM. A robust privacy-preserving federated learning model against model poisoning attacks. *IEEE Trans Inf Forensics Secur* 2024;19:6693–708. <https://doi.org/10.1109/TIFS.2024.3420126>.
- [15] Nabavirazavi S, Taheri R, Iyengar SS. Enhancing federated learning robustness through randomization and mixture. *Future Gener Comput Syst Sep*. 2024;158:28–43. <https://doi.org/10.1016/j.future.2024.04.009>.
- [16] Omer AR, Khan MS, Yousafzai A. Robust federated learning: Defence against model poisoning using mean filtering. In: *2024 Horizons of Information technology and Engineering (HITE)*. IEEE; Oct. 2024. p. 1–5. <https://doi.org/10.1109/HITE63532.2024.10777143>.
- [17] Xia F, Cheng W. A survey on privacy-preserving federated learning against poisoning attacks. *Clust Comput Dec*. 2024;27(10):13565–82. <https://doi.org/10.1007/s10586-024-04629-7>.
- [18] Tian Z, Cui L, Liang J, Yu S. A comprehensive survey on poisoning attacks and Countermeasures in machine learning. *ACM Comput Surv Aug*. 2023;55(8):1–35. <https://doi.org/10.1145/3551636>.
- [19] Yaacoub J-PA, Noura HN, Salman O. Security of federated learning with IoT systems: issues, limitations, challenges, and solutions. *Internet Things Cyber-Physical Syst* 2023;3:155–79. <https://doi.org/10.1016/j.iotcps.2023.04.001>.
- [20] Mohy-Eddine M, Guezzaz A, Benkirane S, Azrou M, Farhaoui Y. An ensemble learning based intrusion detection model for industrial IoT security. *Big Data Min Anal Sep*. 2023;6(3):273–87. <https://doi.org/10.26599/BDMA.2022.9020032>.
- [21] Alotaibi Y, Ilyas M. Ensemble-learning framework for intrusion detection to enhance Internet of Things' devices security. *Sensors Jun*. 2023;23(12):5568. <https://doi.org/10.3390/s23125568>.
- [22] Tomer V, Sharma S. Detecting IoT attacks using an ensemble machine learning model. *Futur Internet Mar*. 2022;14(4):102. <https://doi.org/10.3390/fi14040102>.
- [23] Bag S. A Beginners Guide to federated learning. <https://www.analyticsvidhya.com/blog/2021/05/federated-learning-a-beginners-guide/>. [Accessed 29 January 2025].
- [24] K. Martineau, "What is federated learning?" Accessed: January. 29, 2025. [Online]. Available: <https://research.ibm.com/blog/what-is-federated-learning>.
- [25] Zhang C, Xie Y, Bai H, Yu B, Li W, Gao Y. A survey on federated learning. *Knowl Base Syst Mar*. 2021;216:106775. <https://doi.org/10.1016/j.knosys.2021.106775>.
- [26] Pintelas P, Livieris IE. Special issue on ensemble learning and applications. *Algorithms Jun*. 2020;13(6):140. <https://doi.org/10.3390/a13060140>.
- [27] Rezaei A. Using ensemble learning technique for detecting Botnet on IoT. *SN Comput Sci May* 2021;2(3):148. <https://doi.org/10.1007/s42979-021-00585-w>.
- [28] Hazman C, Guezzaz A, Benkirane S, Azrou M. IIDS-SIoEL: intrusion detection framework for IoT-based smart environments security using ensemble learning. *Clust Comput Dec*. 2023;26(6):4069–83. <https://doi.org/10.1007/s10586-022-03810-0>.
- [29] Cao Y, Wang Z, Ding H, Zhang J, Li B. An intrusion detection system based on stacked ensemble learning for IoT network. *Comput Electr Eng Sep*. 2023;110:108836. <https://doi.org/10.1016/j.compeleceng.2023.108836>.
- [30] Ben Saad S, Brik B, Ksentini A. Toward securing federated learning against poisoning attacks in Zero Touch B5G networks. *IEEE Trans Netw Serv Manag Jun*. 2023;20(2):1612–24. <https://doi.org/10.1109/TNSM.2023.3278838>.
- [31] Liu J, Li X, Liu X, Zhang H, Miao Y, Deng RH. DefendFL: a privacy-preserving federated learning scheme against poisoning attacks. *IEEE Transact Neural Networks Learn Syst* 2024;1–14. <https://doi.org/10.1109/TNNLS.2024.3423397>.
- [32] Zhou J, et al. A differentially private federated learning model against poisoning attacks in edge computing. *IEEE Trans Dependable Secure Comput* 2022;1. <https://doi.org/10.1109/TDSC.2022.3168556>.
- [33] Chen X, Yu H, Jia X, Yu X. APFed: Anti-poisoning attacks in privacy-preserving heterogeneous federated learning. *IEEE Trans Inf Forensics Secur* 2023;18:5749–61. <https://doi.org/10.1109/TIFS.2023.3315125>.
- [34] Zhang Z, Zhang Y, Guo D, Yao L, Li Z. SecFedNIDS: robust defense for poisoning attack against federated learning-based network intrusion detection system. *Future Gener Comput Syst Sep*. 2022;134:154–69. <https://doi.org/10.1016/j.future.2022.04.010>.
- [35] Zavad S, et al. FedCust: Offloading hyperparameter customization for federated learning. *Perform Eval Mar*. 2025;167:102450. <https://doi.org/10.1016/j.peva.2024.102450>.
- [36] Jayalatchumy D, Ramalingam R, Balakrishnan A, Safran M, Alfarhood S. Improved Crow search-based feature selection and ensemble learning for IoT intrusion detection. *IEEE Access* 2024;12:33218–35. <https://doi.org/10.1109/ACCESS.2024.3372859>.
- [37] Jeffrey N, Tan Q, Villar JR. Using ensemble learning for anomaly detection in Cyber-Physical systems. *Electronics Apr*. 2024;13(7):1391. <https://doi.org/10.3390/electronics13071391>.
- [38] Osman M, He J, Zhu N, Mokbal FMM. An ensemble learning framework for the detection of RPL attacks in IoT networks based on the genetic feature selection approach. *Ad Hoc Netw Jan*. 2024;152:103331. <https://doi.org/10.1016/j.adhoc.2023.103331>.
- [39] Fares H, Zeroual M, Karim A, Maleh Y, Baddi Y, Aknin N. Machine learning, deep learning and ensemble learning based approaches for intrusion detection enhancement. *EDPACS Jan*. 2025;70(1):31–51. <https://doi.org/10.1080/07366981.2024.2422645>.
- [40] Caldas S, et al. LEAF: a benchmark for federated settings. <http://arxiv.org/abs/1812.01097>; Dec. 2018.
- [41] Moustafa N, Slay J. UNSW-NB15: a comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set). In: *2015 military communications and information systems conference (MilCIS)*. IEEE; Nov. 2015. p. 1–6. <https://doi.org/10.1109/MilCIS.2015.7348942>.
- [42] S. Huang, "KDD Cup 1999 Data." [Online]. Available: <https://www.kaggle.com/datasets/galex/h/kdd-cup-1999-data>.
- [43] Mohi-ud-din G. NSL-KDD. IEEE Dataport 2018. <https://doi.org/10.21227/425a-3e55>.
- [44] Garcia-Gonzalez D, Rivero D, Fernandez-Blanco E, Luaces MR. A public domain dataset for real-life human activity recognition using smartphone sensors. *Sensors (Switzerland)* 2020. <https://doi.org/10.3390/s20082200>.
- [45] Neto ECP, Dadkhah S, Ferreira R, Zohourian A, Lu R, Ghorbani AA. CICIOT2023: a real-time dataset and benchmark for large-scale attacks in IoT environment. *Sensors Jun*. 2023;23(13):5941. <https://doi.org/10.3390/s23135941>.
- [46] He Haibo, Bai Yang, Garcia EA, Li Shutao. ADASYN: adaptive synthetic sampling approach for imbalanced learning. In: *2008 IEEE international joint conference on neural networks (IEEE world congress on computational intelligence)*. IEEE; Jun. 2008. p. 1322–8. <https://doi.org/10.1109/IJCNN.2008.4633969>.
- [47] Google Colab." Accessed: January. 29, 2025. [Online]. Available: <https://colab.research.google.com/>.
- [48] NumPy." [Online]. Available: <https://numpy.org/>.
- [49] pandas." Accessed: January. 29, 2025. [Online]. Available: <https://pandas.pydata.org/>.
- [50] "scikit-learn." Accessed: January. 29, 2025. [Online]. Available: <https://scikit-learn.org/stable/>.
- [51] Matplotlib." [Online]. Available: <https://matplotlib.org/>.
- [52] "seaborn", [Online]. Available: <https://seaborn.pydata.org/>.
- [53] Prathiba SB, et al. Fortifying federated learning in IIoT: leveraging blockchain and digital twin innovations for enhanced security and resilience. *IEEE Access* 2024;12:68968–80. <https://doi.org/10.1109/ACCESS.2024.3401039>.
- [54] Wasilewska M, Bogucka H. "Protection against poisoning attacks on federated learning-based spectrum sensing \$ \setminuslg \$ } ?>,". *IEEE J Sel Area Commun June* 2025;43(6):2042–55. <https://doi.org/10.1109/JSAC.2025.3560285>.
- [55] Shashidhar Ryushan et al., "DBFFL-defending against dynamic poisoning attacks in federated learning in 5G and beyond systems", in *The proceedings of 2025 IEEE international conference on communications (ICC)*: Montreal, Canada.
- [56] Wan Tao, Feng Shun, Liao Weichuan, Jiang Nan, Zhou Jie. A secure and fair client selection based on DDPG for federated learning. *Int J Intell Syst* 2024;2314019. <https://doi.org/10.1155/2024/2314019>. 14 pages, 2024.